Emotion

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Affect Dynamics in Adolescent Depression: Are All Equilibria Worth Returning to?

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Difficulties in emotion regulation during adolescence have been linked to depression. Early identification of deficits in emotion regulation may help prevent the onset of depression. This study investigated whether emotion regulation dynamics, particularly the strength of regulation to one's affective equilibrium (damping) inferred from the damped oscillator model, predicts future depressive symptoms in adolescents. We hypothesized that the relationship between damping and long-term outcomes would depend on the position of an individual's equilibrium. From July 2019 to May 2024, participants (N = 115) aged 12–15 completed smartphone-based ecological momentary assessment for 30 days, rating six emotions four times daily. The damped oscillator model was applied to each participant's time-series data, yielding person-specific frequency of oscillations (η) and damping (ζ) parameters. These parameters were then used to predict depressive symptoms at 6- and 12-month follow-ups controlling for baseline levels of depression. Results revealed that the interaction between the damping parameter and the equilibrium position significantly predicted depressive symptoms for sadness, but not for other emotions. For individuals with higher equilibrium levels of sadness, stronger damping predicted higher follow-up depressive symptoms. Additionally, higher frequency of oscillation around the equilibrium—representing greater elasticity and less rigidity—of two emotions (interest and happiness) predicted fewer depressive symptoms. These findings suggest that the adaptive value of rapid emotional recovery depends on one's baseline emotional state. Tracking emotion regulation dynamics of both positive and negative emotions may improve our ability to identify adolescents at risk for depression before symptoms emerge, thereby informing targeted intervention and prevention efforts.

Keywords: adolescence, emotion dynamics, emotion regulation, depression

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The prevalence of depression among adolescents has surged substantially over the past 2 decades (Twenge et al., 2019). Evidence points to further increases in child and adolescent depression symptoms during the COVID-19 pandemic (Madigan et al., 2023), contributing to what is now described as a mental health epidemic among teens. Early identification of adolescents at risk of developing depression is crucial for timely intervention and prevention of this very often recurrent and debilitating disorder. Deficits in emotion regulation have been identified as an important risk and maintenance factor for depression, particularly during adolescence (Schäfer et al., 2017). This developmental stage is characterized by considerable developmental change, especially in emotional functioning and regulation. Adolescents experience more extreme emotional states, with wider fluctuations between high and low moods than adults and

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children experience (Maciejewski et al., 2015; Riediger & Klipker, 2014; Riediger & Luong, 2015), yet this is also when their regulatory abilities are rapidly developing (Hollenstein & Lanteigne, 2018). While increased emotional intensity, reactivity, and variability are commonly observed during this period (for a review, see Bailen et al., 2019; Rosenblum & Lewis, 2006), some adolescents exhibit more extreme mood fluctuations and greater instability than others, and individual differences in these patterns have been linked to risk for psychopathology (Hollenstein & Lougheed, 2013; Maciejewski et al., 2019). Therefore, identifying emotion regulatory processes that differentiate adolescents at risk for depression from those on more adaptive developmental paths offers a timely opportunity for targeted intervention and prevention.

Emotion regulation is fundamentally a process that unfolds over time, and therefore it should be theoretically and empirically modeled as a dynamic temporal process. Theoretical models, such as Larsen's (2000) mood regulation framework, conceptualize mood regulation through homeostatic principles, where individuals tend to return to a baseline emotional state (i.e., an equilibrium) after deviating from it. The fluctuations in emotional states around this equilibrium provide insights into how individuals regulate their emotions, with faster returns to equilibrium after short-term perturbations—such as a disagreement with a friend and adaptation to long-term persistent challenges, like ongoing academic pressure—reflecting the system's ability to recover and regulate effectively. This framework conceptualizes emotion regulation like a thermostat. When there is a discrepancy between one's baseline emotional state and the current state, a natural homeostatic mechanism kicks in to reduce these discrepancies, bringing the individual back to their baseline (Hollenstein & Lanteigne, 2018; Larsen, 2000). For example, individuals with high levels of neuroticism, due to their heightened sensitivity to negative stimuli and lower regulation abilities, may experience more pronounced emotional fluctuations and a slower return to equilibrium (Larsen & Ketelaar, 1991).

To test hypotheses related to mood regulation models, researchers have adopted the damped oscillator model (DOM; Boker & Graham, 1998). The DOM is a specific application of dynamic system theory that provides a direct mathematical foundation for this purpose. While dynamic system theory offers a broad framework for understanding time-varying systems, the DOM focuses on the homeostatic regulation of emotions at the idiographic (within-person) level. This involves tracking how emotional states deviate from, oscillate around, and ultimately return to a baseline or equilibrium (for a comprehensive overview of this model, refer to Ollero et al., 2023). By applying the DOM, researchers can mathematically model person-specific regulatory processes in affect, capturing the patterns of mood fluctuations and stabilization over time. Beyond standard metrics like mean and variability in emotions, the DOM highlights two critical aspects of emotional dynamics: *frequency*, how fast the system oscillates around its equilibrium point following a perturbation, and *damping*, the reduction in amplitude of these oscillations allowing the system to return to equilibrium (Bisconti et al., 2004; Ollero et al., 2023). Both frequency and damping are appealing ways to conceptualize aspects of mood regulation or dysregulation. The damping parameter offers a particularly novel way to assess emotional recovery, indicating how effectively an individual can return to a stable affective state after experiencing a negative event. As such, damping is considered an affective regulation or resilience parameter (Bergeman et al., 2021).

A number of previous studies have used the DOM with affective states to examine aspects of affect regulation. Chow et al. (2005) introduced the DOM for formally studying affect regulation and demonstrated individual differences in the frequency of oscillations and damping. Similarly, Hoeksma et al. (2007) found significant oscillations and damping of anger in a sample of 30 participants aged 10–13, demonstrating that in this age group anger rises rapidly, levels off slowly, and returns to a baseline. Montpetit et al. (2010) found significant damping of negative affect (NA) among aging adults who completed daily measures of NA and perceived stress for 56 days. In addition, individuals higher in trait resilience showed a quicker return to equilibrium. Further analyses revealed that people with greater damping in perceived stress (i.e., faster return to stress equilibrium) had fewer health problems 3–5 years later, supporting the criterion and predictive validity of the damping parameter (Bergeman et al., 2021). However, a comparable mathematical model for affective dynamics (the DynAffect Model; Oravecz et al., 2011) found mixed results: While more negative affective equilibrium and greater deviation from the equilibrium were associated with borderline personality disorder, a disorder characterized by emotion dysregulation, the speed of return to equilibrium showed less consistent results, with slower return to equilibrium differentiating borderline personality disorder from healthy controls in only one of three data sets included in the study (Ebner-Priemer et al., 2015).

A possible source of the inconsistent results in the predictive value of speed of return to equilibrium is that not all equilibria are adaptive. For example, if an individual has a high baseline level of NA, a higher damping to this equilibrium position is likely maladaptive. In such a dynamical system, regulation toward this high NA equilibrium would imply *dysregulation*. Hence, faster return to equilibrium can be considered a metric of affective regulation only when the equilibrium is at a desirable or healthy level. Conversely, a quicker return to elevated levels of NA would serve to maintain high NA—a significant risk factor for depression (Scheffer et al., 2024a). A similar interaction has been observed with other emotion dynamics. Specifically, in a large-scale analysis of seven data sets, Maciejewski et al. (2023) found that higher variability in NA was associated with more depressive symptoms among individuals with low NA, but with fewer symptoms among those with very high NA. Therefore, when investigating the predictive value of return to equilibrium, it is imperative to consider the position of the equilibrium. This approach may allow a more accurate interpretation of damping effects in the context of emotional regulation and mental health outcomes.

Accordingly, the present study assessed whether damping (the strength of regulation to equilibrium) inferred from the DOM is predictive of depressive symptoms in the long term in a sample of adolescents including those at risk of depression. Importantly, we assessed whether the relationship between regulation and long-term outcomes depends on an individual's equilibrium position. We hypothesized that the association between damping and future depressive symptoms would be moderated by the equilibrium position such that higher damping in NA would be negatively associated with future depression for adolescents with lower baseline NA but positively associated with depression for those with high baseline NA. Conversely, higher damping in positive affect

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(PA) would be negatively associated with future depression for adolescents with higher baseline PA but positively associated with depression for those with low baseline PA. Importantly, we assessed whether these dynamic parameters offer any incremental predictive value relative to commonly used metrics in intensive longitudinal studies: the person-specific mean and standard deviation in affect.

Method

Participants

The initial sample included 115 adolescents ages 12–15 (58 female, 57 male; $M_{age} = 13.5$, $SD = 1.05$), recruited from the greater Boston area between July 2019 and May 2024. This study examined adolescents' risk for developing depression based on having a biological parent with a history of depression. At-risk participants had a biological parent meeting *Diagnostic and Statistical Manual of Mental Disorders,* Fifth Edition criteria for major depressive disorder (MDD; current or past). Healthy controls had no first-degree relatives with MDD, generalized anxiety disorder, bipolar disorder, or a psychotic disorder. Participants on psychotropic medication were excluded. Sixty-eight participants were healthy controls, and 47 were at elevated risk for depression. Additional details on the study inclusion and exclusion criteria and the sample's demographic and baseline characteristics are provided in the Supplemental Table S1. The rate of compliance was quite high, with a mean number of completed ecological momentary assessment (EMA) surveys of 95.72, a median of 103.5, a mode of 121, and a range of 91. There are not enough studies using this type of analysis to provide reliable effect size estimates, and previous research exploring other dynamics has shown inconsistent results, making it difficult to derive reliable effect sizes. Therefore, we refrained from conducting an a priori power analysis and instead determined our sample size based on those used in prior studies (Nelson et al., 2020).

Procedure

The study was approved by the Mass General Brigham Institutional Review Board (Protocol No. 2018P001486). Participants completed a baseline session either in-person or over Zoom, where they underwent a semistructured clinical interview, the Kiddie Schedule for Affective Disorders and Schizophrenia (Kaufman et al., 1997), and completed online self-report questionnaires on Redcap which included the Center for Epidemiological Studies Depression Scale (Radloff, 1977). Participants then installed the MetricWire app on their smartphones in order to complete 30 days of EMA. EMA surveys were delivered four times per day (at randomly selected times) and asked about participants' current emotions. Follow-up assessments using the baseline self-report measures and Kiddie Schedule for Affective Disorders and Schizophrenia were conducted at 6, 12, 18, and 24 months via Zoom. Recruitment for the present study is ongoing, and fewer participants have completed the 18- and 24-month assessments. Thus, we only used the 6- and 12-month follow-up data in this analysis.

Measures

EMA

Six adjectives (three positive and three negative) were chosen from the Positive and Negative Affect Scale for Children (Laurent et al., 1999). Participants rated their current PA and NA indicating the extent to which they were feeling "happy," "interested," and "excited" (for PA) and "sad," "angry," and "nervous" (for NA) immediately before receiving the survey. Ratings were on a slider, ranging from 0 (*not at all*) to 100 (*as much as possible*).

The Center for Epidemiological Studies Depression Scale is a 20-item self-report measure of depression severity, used at baseline and follow-ups (Radloff, 1977). It evaluates the frequency of depressive symptoms over the past week on a 4-point Likert scale (1 = *rarely or none of the time* to 4 = *most or all of the time*). A higher score indicates a higher severity of depressive symptoms, with four items being reversed scored. Cronbach's α for internal consistency was α = .92.

Data Analysis

We modeled the regulatory process in each of the six affect variables (sad, angry, nervous, happy, interested, and excited) using the damped linear oscillator model. As shown in Equation 1, this model is described by the following second order differential equation:

$$\ddot{x}_t = \eta x_t + \zeta \dot{x}_t, \tag{1}$$

where x_t represents the latent variable describing the position of the system at time t, $\dot{x}_t$ is the velocity or the rate of change, and $\ddot{x}_t$ is the acceleration or rate of change of the velocity. The parameters of interest are η (eta), the frequency parameter, and ζ (zeta), the damping parameter. Together, these parameters describe the movement of a given affective state around an individual's equilibrium. A negative value of η indicates a stable equilibrium, meaning that when the system deviates from the equilibrium, it tends to regulate back toward it, with the magnitude of η describing the frequency of oscillations. The η parameter is analogous to the stiffness or elasticity of a spring: The more the spring is stretched, the faster it returns to equilibrium. If the system is undamped (i.e., $\zeta = 0$), the spring will oscillate around the equilibrium indefinitely, with the frequency of oscillations corresponding with the magnitude of η. A positive value of η indicates that if the system deviates from the equilibrium, it tends to accelerate further away, leading to an unstable system. In contrast, a negative η suggests that the system is self-regulating, with the magnitude of η corresponding to the frequency of oscillations around the equilibrium.

In an undamped system, the spring parameter is expected to oscillate around the equilibrium indefinitely. This is unlikely for psychological processes including affect, which presumably settle at the equilibrium state eventually. The ζ or damping parameter describes the *friction* in the spring or the reduction in the amplitude of oscillations over time. A negative value of ζ indicates the presence of damping in the system. A more negative ζ indicates stronger damping in the system that serves to absorb shocks and restore the system back to equilibrium. The value of ζ can therefore be taken as a measure of regulatory strength or resilience of a system (Boker et al., 2024). To aid readability, the parameters η and ζ will be referred to as spring and damping parameters.

In this study, we fit the DOM using stochastic differential equations (SDE) within a continuous-time framework using the R package *dynr* (Ou et al., 2019). SDE includes both *process* noise and *measurement* noise. Whereas measurement noise captures measurement error in the observed variable, process noise reflects true fluctuations in the underlying latent state of the variable over time.

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Step 1 involved fitting a series of idiographic continuous-time SDE models separately for each individual and for each affective variable. Prior to this, each time series was linearly detrended to meet the assumption of a stationary equilibrium, and only series with nonzero intraindividual variance were retained for modeling, as the model assumes variability in the process being estimated. The resulting residuals were then used in SDE analysis. For SDE models, the continuous measure of time elapsed since the first measurement occasion was scaled by 24 hr. This step gives a larger window of oscillations and better convergence. See the code for details of model fitting (https://osf.io/2pruy/). The idiographic SDE models yielded person-specific values for η (stiffness of the spring) and ζ (damping in the oscillations) parameters for each variable. As a result, for each individual, we obtained six values for η and six for ζ (one for each affective variable). Parameter values from nonconverging models were excluded. Step 2 involved conducting between-persons moderated regression analyses to determine whether these dynamic parameters predicted depression (Center for Epidemiological Studies Depression) at 6- and 12-month follow-ups. These analyses controlled for gender, baseline depression, and the person-specific mean and standard deviation from the time series for each variable. The equilibrium position, or person-specific mean, was included as a moderator.

Transparency and Openness

We report how we determined our sample size, all data exclusions, all manipulations, and all measures in the study, and the study follows the Journal Article Reporting Standards (Kazak, 2018). The data are shared with the National Institute of Mental Health Data Archive. Analysis code and research materials are available at https://osf.io/2pruy/ (Fisher, 2024). This study's design and its analysis were not preregistered.

Results

The idiographic DOM models converged for varying number of individuals across affect variables ranging from 52 (*for anger*) to 63 (*for happy*). A likely reason for nonconvergence is missing data and limited intraindividual variance in some of the time series. All individuals had a negative value for η for all affective variables, indicating self-regulation to equilibrium after perturbation. With the exception of one to two individuals with nonnegative values, all had a negative value for ζ for all variables indicating damping or resilience in all affective systems. Yet, there were considerable individual differences in both these parameters. See Figure 1 for examples of varying levels of spring and damping, with more negative values indicating stronger spring and damping, at different levels of equilibria in sadness. See Supplemental Table S2 for detailed distribution of the DOM parameters.[1]

In the between-person regression analyses, baseline depressive symptoms and being female predicted change in depressive symptoms at 6- and 12-month follow-up, as expected (see Tables 1–4). As shown in Table 1, overall, the spring parameter (η) and its interaction with the person-specific equilibrium position did not predict change in follow-up depressive symptoms for NA variables (sad, angry, nervous). We found significant main effects for the spring parameter (Table 3): Stronger spring (more negative η values) in *happy* and *interested* predicted greater reductions in depressive symptoms at 6-month follow-up, controlling for gender, baseline depression, and person-specific mean and standard deviation of the corresponding affect. For *interested*, the spring parameter substantially improved the base model, increasing explained variance by 19% points at 6 months. For *happy*, while the spring parameter was significant at both 6- and 12-month follow-ups, its incremental contribution was modest (1% at 6 months, 7% at 12 months) beyond the base model. With Benjamini–Hochberg corrected *p* values, the effect for *Happy* was no longer significant while the spring for Interested continued to significantly predict depression at 6-month follow-up (see Supplemental Materials for the full table).

In line with our hypothesis, there was a significant interaction effect between the damping parameter (ζ) and the equilibrium position (indicated by mean affect) for sadness (see Table 2). Specifically, for individuals with a higher equilibrium position for sadness, stronger damping (more negative ζ values) predicted increase in depressive symptoms at both follow-up points, but this was not the case for lower equilibrium positions for sadness (see Figure 2). These effects remained significant when controlling for gender, baseline depressive symptoms, and the person-specific mean and standard deviation for sadness. The interaction between damping and the equilibrium accounted for an additional 15%–20% of the variance in follow-up depression scores. Notably, this interaction effect was only observed for sadness, a prominent negative emotion linked to depression and one of the two criterial symptoms of MDD (i.e., sadness and anhedonia). These effects remained significant with Benjamini–Hochberg corrected *p* values. It is also important to mention that the sample size for the regression models decreased substantially once the spring and damping parameters were added to the base model for each variable. This reduction in sample size, due to missing data from nonconvergence in the DOM for several individuals, may have resulted in a loss of power in the regression models. Finally, we repeated our analyses and included the presence of COVID-19 and risk group as covariates. The presence of COVID-19 was determined based on whether any part of a participant's data collection occurred during the local pandemic restriction period (March 2020 to March 2021). These adjustments did not change the pattern of results (Supplemental Tables S3–S6).

Discussion

This study aimed to investigate whether emotion dynamics inferred from the DOM could predict increase in future depressive symptoms in adolescents. By using intensive longitudinal data to model regulatory processes, this study offers a robust alternative to studies using traditional self-reported measures of emotion regulation (i.e., asking participants to retrospectively report on their past emotional states and regulation abilities), aligning with recent calls for more objective and process-oriented approaches to emotion regulation (Kuppens & Verduyn, 2017).

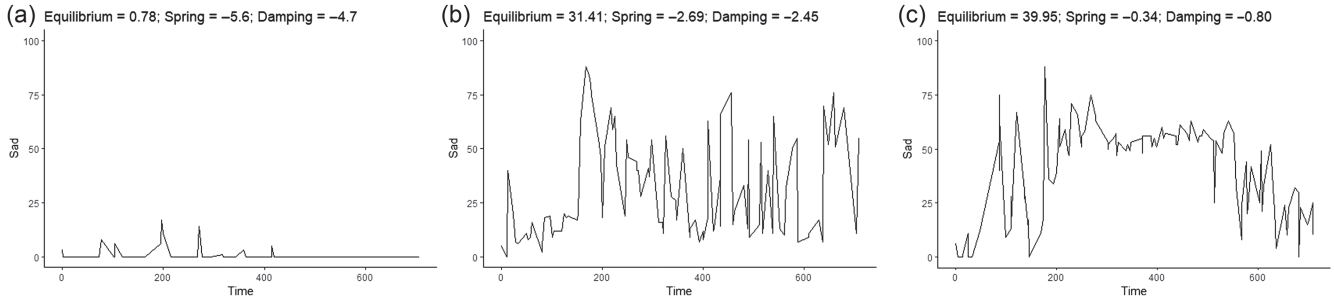
Supporting our hypothesis, we found significant interaction effects between damping and equilibrium position for sadness, but not

[1] We have included the descriptive statistics for *Q*-factor and damping ratio in the Supplemental Materials. The damping ratio quantifies the relationship between damping and oscillation frequency, with higher values indicating faster stabilization with fewer oscillations. The *Q*-factor corresponds with the number of oscillations before the system returns to equilibrium, with higher values indicating more sustained fluctuations.

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Figure 1

Example Plots of Regulatory Dynamics in Sadness



Note. This figure represents example idiographic plots for varying values of dynamic parameters indicating different strength of regulation at different levels of equilibria for sadness. The unit of time is the number of hours elapsed since first measurement. Panel A shows the relatively most regulated dynamic system, which tends to restore to the equilibrium position of 0.78 quickly. Panel B has a weaker regulation than the example in Panel A but tends to maintain the state of the system at the high equilibrium position of 31. Panel C represents weak regulation, and the system often stays away from the equilibrium position of 39.95.

other emotions. Specifically, for individuals with higher sadness equilibrium levels, stronger damping predicted an increase in depressive symptoms at both follow-up timepoints, as it serves to maintain the high sadness equilibrium. These findings suggest that the adaptive value of quick emotional recovery depends on the baseline emotional state, which aligns with the perspective that not all equilibria are adaptive (Hollenstein & Lanteigne, 2018). Together with previous findings showing that the link between NA variability and depressive symptoms depends on individuals' baseline affect levels, our results suggest that the impact of emotion dynamics on depressive symptoms may be contingent on baseline emotional state (Maciejewski et al., 2023).

From a dynamical systems perspective, higher damping indicates greater resilience, reflecting a system's capacity of sustained regulation back to its original equilibrium after perturbation. However, resilience is not inherently adaptive; it can be either beneficial or detrimental depending on the equilibrium position (Scheffer, 2009). If the equilibrium is at a healthy state, maintaining resilience is crucial for preventing a shift toward an unhealthy state. Conversely, if the equilibrium is at an unhealthy state, reducing resilience may be necessary to facilitate a transition to a healthier state (Scheffer et al., 2024b). Therefore, for an individual whose equilibrium position for sadness is high, stronger damping or resilience may serve to maintain elevated sadness levels, potentially leading to detrimental

Table 1

Dynamics (η) of Sadness, Anger, and Nervousness Predicting Depressive Symptoms at 6 and 12 Months

	Sad (N = 52)		Angry (N = 54)		Nervous (N = 60)	
	CESD at 6-month	CESD at 12-month	CESD at 6-month	CESD at 12-month	CESD at 6-month	CESD at 12-month
Variable	*B* (*SE*)	*B* (*SE*)	*B* (*SE*)	*B* (*SE*)	*B* (*SE*)	*B* (*SE*)
Base model						
Gender	−3.08 (1.52)*	−4.66 (1.75)**	−3.65 (1.43)*	−4.84 (1.66)**	−3.71 (1.52)*	−5.66 (1.80)**
Baseline CESD	0.49 (0.11)***	0.64 (0.14)***	0.53 (0.11)***	0.70 (0.14)***	0.56 (0.11)***	0.71 (0.13)***
Mean affect	0.07 (0.10)	0.12 (0.12)	−0.03 (0.13)	−0.04 (0.14)	−0.03 (0.08)	0.06 (0.09)
SD affect	0.12 (0.15)	0.03 (0.17)	0.17 (0.16)	0.20 (0.18)	0.10 (0.12)	−0.13 (0.14)
	R^2 = .32	R^2 = .40	R^2 = .31	R^2 = .40	R^2 = .30	R^2 = .39
η main effect						
Gender	−1.93 (2.50)	−4.07 (3.06)	−2.49 (2.11)	−5.57 (2.48)*	−3.74 (2.34)	−6.03 (2.68)*
Baseline CESD	0.37 (0.16)*	0.65 (0.20)**	0.43 (0.23)†	0.80 (0.36)*	0.51 (0.14)***	0.66 (0.17)***
Mean affect	0.09 (0.15)	0.22 (0.17)	0.03 (0.21)	0.22 (0.28)	−0.04 (0.12)	0.07 (0.13)
SD affect	0.45 (0.23)†	0.06 (0.28)	0.04 (0.23)	−0.10 (0.29)	0.15 (0.18)	−0.18 (0.20)
η	−0.79 (0.72)	0.61 (0.93)	0.69 (0.62)	0.62 (0.76)	0.28 (0.72)	0.54 (0.73)
	R^2 = .31	R^2 = .38	R^2 = .11	R^2 = .32	R^2 = .30	R^2 = .36
η interact						
Gender	−2.67 (2.62)	−4.62 (2.99)	−2.59 (2.11)	−5.56 (2.49)*	−3.65 (2.44)	−6.25 (2.80)*
Baseline CESD	0.44 (0.18)*	0.78 (0.21)***	0.42 (0.23)†	0.80 (0.36)*	0.51 (0.14)***	0.67 (0.18)***
Mean affect	0.20 (0.19)	0.44 (0.21)	0.16 (0.25)	0.34 (0.31)	−0.06 (0.15)	0.10 (0.16)
SD affect	0.43 (0.23)	0.06 (0.27)	0.06 (0.24)	−0.05 (0.30)	0.15 (0.18)	−0.18 (0.21)
η	−1.37 (0.95)	−0.57 (1.16)	0.06 (0.88)	−0.03 (1.03)	0.42 (1.12)	0.23 (0.07)
η × mean affect	0.95 (0.10)	0.20 (0.13)	0.08 (0.08)	0.09 (0.09)	−0.01 (0.07)	0.02 (0.07)
	R^2 = .31	R^2 = .41	R^2 = .11	R^2 = .32	R^2 = .29	R^2 = .35

Note. More negative values of η indicate stronger spring. N for each variable indicates the number of idiographic models converged and hence the effective sample size for the full regression model for that variable. *SE* = standard error; CESD = Center for Epidemiologic Studies Depression Scale.
$^{\dagger}p < .10$. $^{*}p < .05$. $^{**}p < .01$. $^{***}p < .001$.

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Table 2
Dynamics (ζ) of Sadness, Anger, and Nervousness Predicting Depressive Symptoms at 6 and 12 Months

	Sad (N = 52)		Angry (N = 54)		Nervous (N = 60)	
	CESD at 6-month	CESD at 12-month	CESD at 6-month	CESD at 12-month	CESD at 6-month	CESD at 12-month
Variable	*B* (*SE*)	*B* (*SE*)	*B* (*SE*)	*B* (*SE*)	*B* (*SE*)	*B* (*SE*)
Base model						
Gender	−3.08 (1.52)*	−4.66 (1.75)**	−3.65 (1.43)*	−4.84 (1.66)**	−3.71 (1.52)*	−5.66 (1.80)**
Baseline CESD	0.49 (0.11)***	0.64 (0.14)***	0.53 (0.11)***	0.70 (0.14)***	0.56 (0.11)***	0.71 (0.13)***
Mean affect	0.07 (0.10)	0.12 (0.12)	−0.03 (0.13)	−0.04 (0.14)	−0.03 (0.08)	0.06 (0.09)
SD affect	0.12 (0.15)	0.03 (0.17)	0.17 (0.16)	0.20 (0.18)	0.10 (0.12)	−0.13 (0.14)
	R^2 = .32	R^2 = .40	R^2 = .31	R^2 = .40	R^2 = .30	R^2 = .39
ζ main effect						
Gender	−1.73 (2.43)	−2.14 (2.91)	−2.50 (2.13)	−5.71 (2.47)*	−3.74 (2.34)	−5.83 (2.69)*
Baseline CESD	0.35 (0.15)*	0.52 (0.19)*	0.38 (0.24)	0.70 (0.35)*	0.51 (0.13)***	0.70 (0.16)***
Mean affect	0.08 (0.14)	0.29 (0.16)†	0.08 (0.21)	0.25 (0.26)	−0.03 (0.12)	0.08 (0.13)
SD affect	0.48 (0.23)*	0.14 (0.26)	0.05 (0.24)	−0.06 (0.29)	0.15 (0.18)	−0.17 (0.20)
ζ	−1.76 (0.98)†	−2.37 (1.26)†	0.43 (0.72)	0.91 (0.86)	0.74 (1.08)	−0.16 (1.26)
	R^2 = .34	R^2 = .44	R^2 = .10	R^2 = .33	R^2 = .31	R^2 = .35
ζ interact						
Gender	0.01 (2.28)	−0.23 (2.55)	−2.22 (2.12)	−5.28 (2.54)*	−3.47 (2.38)	−6.45 (2.77)*
Baseline CESD	0.28 (0.14)†	0.51 (0.16)**	0.41 (0.24)†	0.74 (0.35)*	0.50 (0.13)***	0.74 (0.17)***
Mean affect	−0.19 (0.16)	−0.05 (0.17)	−0.14 (0.27)	0.12 (0.31)	−0.06 (0.16)	0.19 (0.17)
SD affect	0.39 (0.21)†	−0.00 (0.23)	0.03 (0.24)	−0.12 (0.30)	0.15 (0.18)	−0.18 (0.21)
ζ	0.53 (1.18)	1.18 (1.50)	1.56 (1.13)	1.74 (1.35)	1.14 (1.65)	−1.77 (2.11)
ζ × mean affect	−0.29 (0.10)**	−0.38 (0.11)**	−0.11 (0.09)	−0.09 (0.11)	−0.03 (0.08)	0.09 (0.10)
	R^2 = .46	R^2 = .59	R^2 = .11	R^2 = .32	R^2 = .29	R^2 = .35

Note. More negative values of ζ indicate stronger damping. *N* for each variable indicates the number of idiographic models converged and hence the effective sample size for the full regression model for that variable. *SE* = standard error; CESD = Center for Epidemiologic Studies Depression Scale.
† $p < .10$. * $p < .05$. ** $p < .01$. *** $p < .001$.

long-term outcomes, including an increased risk for depression. Given that sadness is a core symptom of depression (American Psychiatric Association, 2013), it is not surprising that stabilization around high sadness over time can reinforce negative emotional states and increase the risk of depression.

For positive emotions (happy and interested), we found that a weaker spring parameter (lower frequency of oscillation) predicted an increase depressive symptoms irrespective of the equilibrium position. Notably, besides sadness, loss of interest (anhedonia) is the second criterial symptom of MDD. A weak spring parameter reflects a lack of elasticity in an affective system as the spring moves sluggishly after getting perturbed and potentially becoming stuck in a displaced position—similar to the concept of emotional inertia or rigidity. In contrast, a stronger spring (higher frequency) reflects a more elastic system with a better capacity to bounce back quickly from emotional perturbations, indicating stronger regulatory abilities (Ollero et al., 2023). For example, following a disappointing test grade, an adolescent with a well-regulated system might show fluctuations in interest levels, decreasing when reminded of the failure but rebounding at other times. In contrast, a less regulated system would show persistent low interest, with minimal fluctuation back toward previous levels. These findings suggest that rigid or poorly regulated positive emotions may contribute to depression risk. These findings corroborate previous research showing that rigidity in positive emotions, measured through emotional inertia, predicts increased depression (Kuppens et al., 2010, 2012), though some studies have found no such relationship (Heininga et al., 2019). Future research should examine oscillation frequency across different emotions and time scales, as excessive and undamped fluctuations might indicate instability or lability. If replicated, our findings suggest interventions should target not only negative emotional patterns but also enhance the regulatory capacity of positive emotions.

Constraints on Generality

The primary limitation of this study was a loss of power due to convergence issues in idiographic analyses, leading to a reduced sample size for between-subjects. Longer time series would increase the convergence rates, resulting in higher sample sizes for between-person analyses. Another limitation of this study is the assumption of a static equilibrium over the 1-month period, which while reasonable in this time frame, may not capture potential changes in emotional baselines. Future studies with longer time series could explore models that allow for changes in the equilibrium position and dynamics. Such time-varying models can also help identify changes in the dynamics over time and whether such changes predict the emergence of depression serving as early warning signals (Tonge et al., 2024). Furthermore, our findings should be interpreted in the light of a specific modeling constraint: The parameters do not distinguish between upward and downward deviations from the equilibrium. Therefore, a stronger damping parameter reflects how strongly a deviation is corrected regardless of the direction of the deviation. As such, it is certainly possible for state sadness to rise above an already high equilibrium, and strong regulation will actually be beneficial and offer momentary relief. However, a strongly damped system will continue to correct for deviations—both positive and negative—and over time maintain the maladaptive equilibrium.

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Table 3

Dynamics (η) of Happy, Interest, and Excitement Predicting Depressive Symptoms at 6 and 12 Months

	Happy (N = 63)		Interested (N = 59)		Excited (N = 61)	
	CESD at 6-month	CESD at 12-month	CESD at 6-month	CESD at 12-month	CESD at 6-month	CESD at 12-month
Variable	*B* (*SE*)	*B* (*SE*)	*B* (*SE*)	*B* (*SE*)	*B* (*SE*)	*B* (*SE*)
Base model						
Gender	−2.86 (1.50)	−4.51 (1.76)*	−3.73 (1.42)*	−5.39 (1.64)**	−3.84 (1.43)**	−5.38 (1.66)**
Baseline CESD	0.49 (0.10)***	0.67 (0.13)***	0.53 (0.10)***	0.68 (0.13)***	0.53 (0.10)***	0.69 (0.13)***
Mean affect	−0.05 (0.04)	−0.05 (0.05)	−0.04 (0.03)	−0.05 (0.03)	−0.03 (0.03)	−0.04 (0.03)
SD affect	0.19 (0.13)	0.11 (0.15)	0.09 (0.10)	−0.06 (0.12)	0.05 (0.09)	−0.20 (0.10)
	R^2 = .34	R^2 = .40	R^2 = .31	R^2 = .40	R^2 = .31	R^2 = .39
η main effect						
Gender	−4.54 (2.44)†	−9.63 (3.03)**	−3.72 (1.97)†	−6.20 (2.72)*	−3.50 (1.85)†	−4.08 (1.95)*
Baseline CESD	0.56 (0.14)***	0.76 (0.19)***	0.68 (0.14)***	0.91 (0.26)**	0.53 (0.11)***	0.68 (0.12)***
Mean affect	−0.08 (0.06)	−0.07 (0.07)	−0.09 (0.04)*	−0.07 (0.05)	−0.04 (0.04)	−0.10 (0.04)*
SD affect	0.00 (0.23)	−0.05 (0.29)	−0.06 (0.13)	−0.17 (0.20)	0.24 (0.12)†	0.07 (0.13)
η	0.70 (0.34)*	0.83 (0.38)*	0.89 (0.38)*	0.51 (0.60)	−0.02 (0.44)	0.53 (0.49)
	R^2 = .34	R^2 = .47	R^2 = .50	R^2 = .41	R^2 = .43	R^2 = .62
η interact						
Gender	−4.78 (2.54)†	−8.93 (3.20)**	−3.61 (1.96)†	−6.17 (2.61)*	−3.50 (1.87)†	−4.34 (1.95)*
Baseline CESD	0.55 (0.14)***	0.78 (0.19)***	0.68 (0.14)**	0.93 (0.25)***	0.54 (0.12)***	0.64 (0.13)***
Mean affect	−0.05 (0.10)	−0.13 (0.12)	−0.13 (0.05)*	−0.19 (0.08)*	−0.04 (0.06)	−0.05 (0.06)
SD affect	0.01 (1.59)	−0.05 (0.29)	−0.04 (0.13)	−0.22 (0.19)	0.24 (0.12)†	0.06 (0.13)
η	0.09 (0.03)	2.22 (1.94)	1.77 (0.83)*	3.81 (1.84)*	0.09 (1.00)	−0.78 (0.03)
η × mean affect	0.01 (0.03)	−0.02 (0.03)	−0.02 (0.02)	−0.07 (0.04)†	−0.00 (0.02)	0.03 (0.03)
	R^2 = .33	R^2 = .46	R^2 = .50	R^2 = .45	R^2 = .42	R^2 = .63

Note. More negative η indicate stronger spring. N for each variable indicates the number of idiographic models converged and hence the effective sample size for the full regression model for that variable. *SE* = standard error; CESD = Center for Epidemiologic Studies Depression Scale.
$^{\dagger}p < .10$. $^{*}p < .05$. $^{**}p < .01$. $^{***}p < .001$.

Relying on self-report measures requires individuals to be aware of their emotional states, but not all emotional experiences are consciously recognized. Future research could address these limitations by incorporating alternative measures, such as passive data from smartphones, text analysis, facial expressions, or auditory data, which can continuously track emotions even if they are not consciously perceived (Ren et al., 2023; Webb et al., 2024). Additionally, we did not explicitly measure the types of stressors or

Table 4

Dynamics (ζ) of Happy, Interest, and Excitement Predicting Depressive Symptoms at 6 and 12 Months

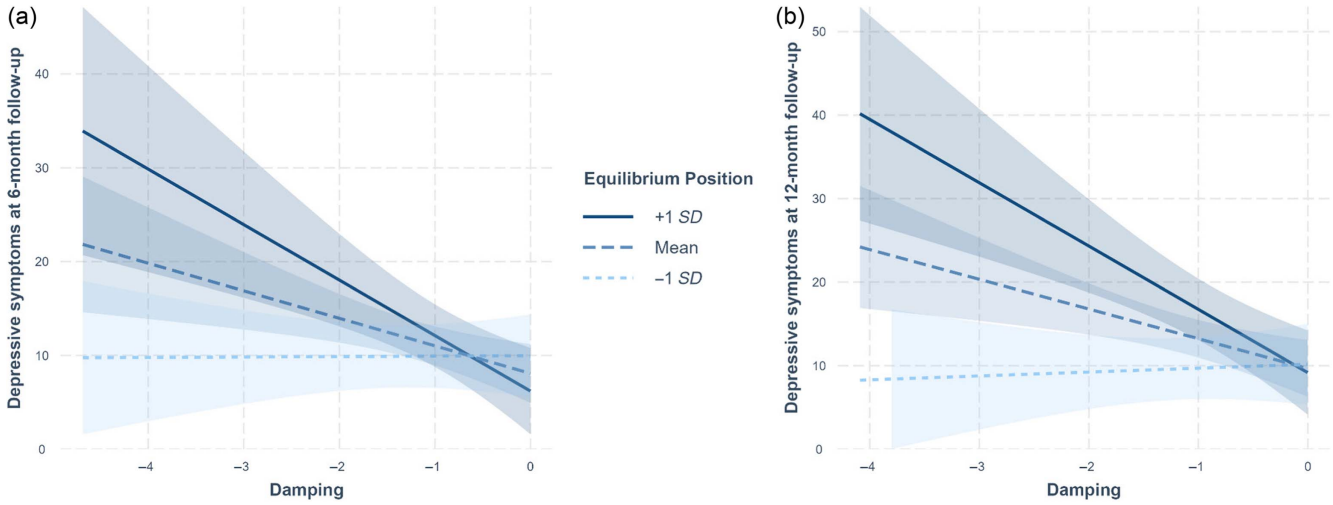
	Happy (N = 63)		Interested (N = 59)		Excited (N = 61)	
	CESD at 6-month	CESD at 12-month	CESD at 6-month	CESD at 12-month	CESD at 6-month	CESD at 12-month
Variable	*B* (*SE*)	*B* (*SE*)	*B* (*SE*)	*B* (*SE*)	*B* (*SE*)	*B* (*SE*)
Base model						
Gender	−2.86 (1.50)	−4.51 (1.76)*	−3.73 (1.42)*	−5.39 (1.64)**	−3.84 (1.43)**	−5.38 (1.66)**
Baseline CESD	0.49 (0.10)***	0.67 (0.13)***	0.53 (0.10)***	0.68 (0.13)***	0.53 (0.10)***	0.69 (0.13)***
Mean affect	−0.05 (0.04)	−0.05 (0.05)	−0.04 (0.03)	−0.05 (0.03)	−0.03 (0.03)	−0.04 (0.03)
SD affect	0.19 (0.13)	0.11 (0.15)	0.09 (0.10)	−0.06 (0.12)	0.05 (0.09)	−0.20 (0.10)
	R^2 = .34	R^2 = .40	R^2 = .31	R^2 = .40	R^2 = .31	R^2 = .39
ζ main effect						
Gender	−3.94 (2.46)	−9.47 (3.40)**	−2.79 (2.10)	−5.16 (2.70)†	−3.62 (1.88)	−4.07 (1.98)*
Baseline CESD	0.57 (0.15)***	0.79 (0.22)**	0.72 (0.15)***	0.92 (0.25)***	0.53 (0.12)***	0.71 (0.13)***
Mean affect	−0.09 (0.07)	−0.08 (0.08)	−0.09 (0.04)*	−0.08 (0.05)	−0.04 (0.04)	−0.10 (0.04)*
SD affect	0.04 (0.24)	−0.1 (0.31)	−0.03 (0.14)	−0.13 (0.20)	0.24 (0.12)†	0.09 (0.13)
ζ	1.13 (0.75)	1.36 (1.05)	1.03 (0.85)	1.55 (1.17)	−0.31 (0.95)	0.74 (0.96)
	R^2 = .31	R^2 = .42	R^2 = .45	R^2 = .42	R^2 = .43	R^2 = .62
ζ interact						
Gender	−3.83 (2.49)	−9.20 (3.39)*	−2.79 (2.11)	−5.28 (2.70)†	−3.70 (1.92)*	−4.39 (2.11)*
Baseline CESD	0.59 (0.15)***	0.83 (0.22)***	0.73 (0.15)***	0.89 (0.25)**	0.52 (0.12)***	0.68 (0.14)***
Mean affect	−0.15 (0.14)	−0.22 (0.17)	−0.12 (0.06)†	−0.01 (0.08)	−0.02 (0.07)	−0.07 (0.07)
SD affect	−0.00 (0.25)	−0.11 (0.33)	−0.02 (0.14)	−0.11 (0.20)	0.24 (0.12)†	0.08 (0.13)
ζ	2.67 (3.14)	5.03 (3.78)	2.02 (1.66)	−0.57 (2.43)	−0.73 (1.91)	−0.06 (1.92)
ζ × mean affect	−0.03 (0.05)	−0.06 (0.06)	−0.03 (0.04)	0.05 (0.05)	0.01 (0.04)	0.02 (0.04)
	R^2 = .30	R^2 = .42	R^2 = .45	R^2 = .42	R^2 = .42	R^2 = .60

Note. More negative values of ζ indicate stronger damping. *SE* = standard error; CESD = Center for Epidemiologic Studies Depression Scale.
$^{\dagger}p < .10$. $^{*}p < .05$. $^{**}p < .01$. $^{***}p < .001$.

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Figure 2

Dynamics in Sadness and Depressive Symptoms at 6-Month (A) and 12-Month (B) Follow-Up



Note. See the online article for the color version of this figure.

emotional triggers during the assessment period, nor did we assess strategies used to regulate emotions back to equilibrium. To improve our understanding of emotion regulation dynamics as described by the DOM, future studies could use intensive experience sampling method to capture emotional states immediately after triggering events and monitor the recovery process (Chow et al., 2005). Combining continuous measures with the assessment of affective triggers would enhance our understanding of individual differences in emotional recovery rates. Finally, in this study, the majority of participants were White, which restricts the generalizability of our findings to more diverse populations. Future research should aim to include participants from a broader range of racial and ethnic backgrounds to enhance the applicability of the results.

Aside from these limitations, this study has several notable strengths. By collecting data multiple times a day, we achieved fine-grained tracking of emotional dynamics, allowing for a nuanced understanding of regulation processes. Our use of sophisticated methods, specifically SDE, directly models movement around the equilibrium, providing a more precise measure of regulation compared to traditional metrics such as person-specific standard deviation and inertia. Additionally, we addressed the interaction between dynamics and the equilibrium position, thereby addressing a major gap in the literature. The study's robustness is further supported by two follow-up time points and by controlling for several relevant baseline and time-series variables, demonstrating that the effects of emotional dynamics are consistent and reliable.

In conclusion, this study advances our understanding of emotion regulation dynamics and their role in depression, emphasizing the importance of considering the interplay between equilibrium position and dynamics when assessing emotional recovery and risk among adolescents. Our findings suggest that monitoring these dynamics could enhance early identification of adolescents at risk for developing depression. These dynamics may be more malleable and amenable to treatment than those observed after the onset of a full depressive episode (Scheffer et al., 2024a), thus providing critical windows for intervention. The specific dynamics observed could also inform targeted intervention or, ideally, prevention strategies to reduce depression risk before episode onset. Future research should continue to explore these dynamic processes, potentially incorporating longer time series, larger samples, and time-varying models. In addition, studies should investigate whether identifying maladaptive emotional dynamics and implementing tailored interventions can lead to changes in these dynamics and lower rates of depression.

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